TSYP 14

IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

GENERAL SPECIFICATIONS BOOK



CHALLENGE

THEME: MINDS DESIGNING RESILIENCE

TSYP 14

IEEE TURKISH STUDENT AND YOUNG PROFESSIONAL CONGRESS

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THE NOOSPHERE CONCEPT & CHALLENGE OBJECTIVE

The term noosphere describes a layer of collective human thought and reasoning: the way humanity's shared ideas shape the world. This challenge borrows that image deliberately. Teams aren't just asked to build a piece of software or hardware, but to act as architects of collective intelligence, helping communities reason, decide, and act wisely when conditions are difficult.

Recent crises, from natural disasters to public health emergencies, show that technology alone rarely saves a system under stress. What usually breaks first is the decision-making around it: fragmented information, eroded trust, and shortcuts taken under pressure.

In one sentence: Architects of the Noosphere challenges student teams to design ethical, collaborative intelligence systems, whether technological, organizational, or social, that strengthen how communities make decisions and recover under pressure.



AOTN

ARCHITECTS OF THE NOOSPHERE

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CHALLENGE SCOPE

AOTN is a progressive innovation journey grounded in sustainability, with challenges framed around the UN Sustainable Development Goals (SDGs).

Teams identify a genuine problem, design a solution, build a working prototype, and defend a complete business case in front of a jury. Along the way, teams build real skills in research, design thinking, technical execution, collaboration, and communication, culminating in a final pitch at the TSYP Congress.

Teams align their project with one of three thematic pillars:

- Sustainable Education
- Sustainable Industry
- Sustainable Agriculture



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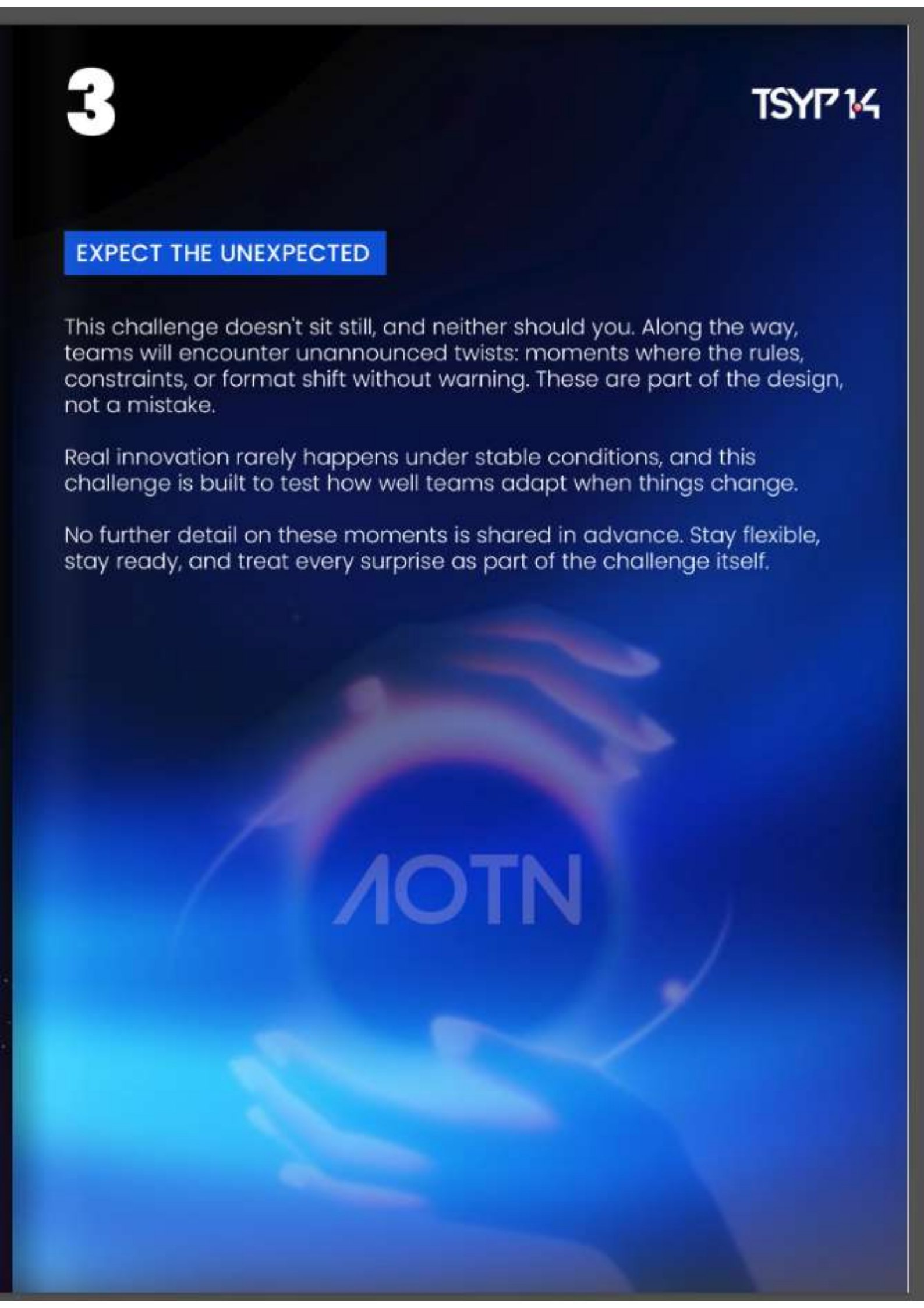
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EXPECT THE UNEXPECTED

This challenge doesn't sit still, and neither should you. Along the way, teams will encounter unannounced twists: moments where the rules, constraints, or format shift without warning. These are part of the design, not a mistake.

Real innovation rarely happens under stable conditions, and this challenge is built to test how well teams adapt when things change.

No further detail on these moments is shared in advance. Stay flexible, stay ready, and treat every surprise as part of the challenge itself.



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CHALLENGE STRUCTURE

The challenge is organized into five phases, each with a clear objective and approximate duration

Phase	Focus	Approx Duration	Elimination
Phase 0	Team formation & alignment	2 weeks	No
Phase 1	Problem exploration	3 weeks	Yes
Phase 2	Solution design & feasibility	3 weeks	Yes
Phase 3	Prototyping	3/4 weeks	Yes
Phase 4	Business, impact & final pitch	2 weeks	No

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Phase 0, Kickoff & Alignment

Objective: Form the team and establish a shared working approach before any substantive work begins.

Description: Teams are introduced to the challenge, finalize their team composition, and align internally on how they will collaborate throughout the journey.

Phase 1, Problem Exploration

Objective: Identify and validate a real, well-scoped problem worth solving.

Description: Teams research a problem space, study the people and systems affected by it, and validate their problem through real engagement with target users or domain experts.

Phase 2, Solution Design & Feasibility

Objective: Design a solution to the validated problem and prove that it can realistically be built.

Description: Teams move from problem to solution, defining what they will build, how it will work, and who it serves, while assessing feasibility across technical and economic dimensions.

Phase 3, Prototyping

Objective: Build and test a working version of the solution to prove that the idea works in practice, not just on paper.

Description: Teams build and test a functional prototype or MVP, incorporating real user feedback. This is the most hands-on phase of the challenge.

Phase 4, Business, Impact & Final Pitch

Objective: Translate the validated, prototyped solution into a complete, fundable, and impactful proposal, culminating in the final pitch.

Description: Finalist teams build out their business model, assess the broader impact of their solution, and prepare to defend their project in front of a panel at the TSYP Congress.